

Yule Cai

Bachelor of Science with Honours (Information Technology) Student | Universiti Utara Malaysia (UUM)

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EDUCATION

Universiti Utara Malaysia (UUM), Kedah, Malaysia — Sept 2025–Present

Bachelor of Science with Honours (Information Technology)

Relevant coursework: Introduction to AI; Programming 1 & 2; Mathematics for IT; Data Structures & Algorithm Analysis; Statistics for IT

RESEARCH INTERESTS

Embodied Intelligence & Robotics • Causal AI & Causal Discovery • Reinforcement Learning • LLM-enabled Planning • Explainable AI

PUBLICATIONS

1. Cai, Y., Zhang, X., Liu, S., Lou, J., & Yuan, C. (2026). Beyond Correlation: A Survey on Causal Inference for Dynamics-Aware Perception and Introspective Decision-Making in Robotics. *Nonlinear Dynamics*, 114(13), 869. DOI: 10.1007/s11071-026-12695-2 [First author]
2. Cai, Y., Tan, W., Wang, Y., Du, X., Sun, T., Ai, Q., Bourouis, S., Alalmaei, S., Shao, W., & Yuan, C. (2026). Toward Edge-Intelligent Consumer Robots: A Decoupled RL-LLM Architecture for Robust Navigation under Partial Observability. *Journal of King Saud University - Computer and Information Sciences*, 38, 470. DOI: 10.1007/s44443-026-00870-9 [First author]
3. Yuan, C., Cai, Y. (co-first), Que, H., Pei, Y., Zhang, X., Xie, J., Zhang, Q., Mu, L., & Qiao, F. (2026). KA-IHO: A Kinematic-Aware Improved Hippo Optimization Algorithm for Collision-Free Mobile Robot Path Planning in Complex Grid Environments. *Sensors*, 26(8), 2416. DOI: 10.3390/s26082416
4. Yuan, C., Cai, Y. (co-first), & Zhang, Z. (2026). Application of SHAP-Based Explainable Machine Learning in Remaining Useful Life Prediction for Aircraft Engine Systems. *International Journal of Machine Learning and Cybernetics*, 17(9). DOI: 10.1007/s13042-026-03246-7
5. Xiao, N., Yuan, C.H., Pei, Y.T., Xue, W., & Cai, Y. (2026). A Study of Artificial Intelligence in Writing Assessment for Secondary School Students. *Educational Studies*, 52(4), 504-526. DOI: 10.1080/03055698.2025.2531969
6. Yang, Q., Chen, J., Yuan, C., Na, L., Chen, W., Sun, J., & Cai, Y. (2025). Spatial Distribution Prediction of Environmental Microbial Communities Based on Deep Learning Combined with MAXENT Model. *Proceedings of SPIE*, 13692, 136923X.

ACADEMIC SERVICE

Peer Reviewer — Springer Nature journals (2026–Present): *International Journal of Information Security* (1); *Scientific Reports* (3); *Discover Artificial Intelligence* (2); *Artificial Intelligence Review* (1); *Discover Robotics* (1); *The Journal of Supercomputing* (1). 9 completed peer reviews across 6 journals.

TECHNICAL SKILLS

Programming: Python; Java

AI & Machine Learning: reinforcement learning; causal inference and discovery; explainable AI; computer vision; LLM-enabled planning and agentic systems

Robotics & Simulation: ROS 2; Gazebo; autonomous navigation; path planning; dexterous manipulation

Research & Engineering: Linux; Git; LaTeX; experimental design; scientific visualization; literature review and manuscript preparation